IntelliTwin: Asymmetric, Condition-Adaptive Conformal Prediction for Industrial Turbofan Predictive Maintenance

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Abstract—Predictive maintenance for industrial gas turbine engines requires timely, risk-calibrated replacement decisions to prevent catastrophic in-flight failures while minimizing premature overhaul costs. Conventional Remaining Useful Life (RUL) prognostic systems rely on point regression models that optimize cycle-level accuracy metrics without quantifying predictive uncertainty or aligning with asymmetric maintenance cost structures.

In this work, we present IntelliTwin, a framework integrating data-driven base predictors (Gradient Boosted Regressors and Long Short-Term Memory networks) with hierarchical, decision-aware conformal prediction. IntelliTwin enforces strict engine-level disjoint data partitioning, causal temporal windowing ($w = 30$, stride 1), equal engine weighting across all evaluations, and condition-stratified conformal calibration. By incorporating an asymmetric loss parameter ($\beta = -\ln 2$), the framework modulates nonconformity scores to reflect the severe cost penalties of component failure relative to preventive replacement.

We formalize an experimental evaluation protocol across NASA C-MAPSS sub-datasets FD001, FD002, and FD004, comparing calibrated lower bounds against four distinct matched-conservatism baselines across three families. Statistical significance is evaluated via paired engine-cluster bootstrap inference ($B = 10{,}000$) with pre-registered practical-effect margins and Bonferroni family-wise error rate control (98.75% confirmatory confidence intervals).

Provisional Status Statement: This document represents the pre-registered methodology and evaluation structure; empirical evaluation on the held-out test splits remains pending experimental execution under stable update gates UPD-02, UPD-03, and UPD-04.

Index Terms—Predictive Maintenance, Conformal Prediction, Remaining Useful Life, Turbofan Engines, Digital Twin, Asymmetric Loss, C-MAPSS.

I. Introduction

In safety-critical cyber-physical systems, such as commercial aviation turbofan engines, unexpected component failure incurs catastrophic operational, financial, and life-safety hazards. Consequently, industrial fleet operators have historically adhered to rigid preventive maintenance schedules, overhauling major subsystems at fixed operational intervals (flight hours or cycles). While preventive scheduling mitigates catastrophic in-flight failure, it imposes severe economic costs by routinely discarding components with substantial remaining physical life [1], [2].

Predictive Maintenance (PdM) leverages onboard multi-sensor telemetry to transition from time-based maintenance to condition-based, opportunistic intervention. Machine learning prognostics aims to estimate the Remaining Useful Life (RUL) of degrading machinery, enabling timely maintenance before failure occurs. In recent years, deep neural architectures—including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) recurrent networks, and gradient-boosted decision trees—have achieved remarkable point-prediction accuracy on standardized benchmarks such as NASA C-MAPSS.

However, deterministic point predictors $\hat{y}$ provide no quantification of predictive uncertainty. In sequential decision environments, a deterministic forecast is fundamentally incomplete: an operator cannot determine whether a predicted life of $\hat{y} = 15$ cycles carries high confidence or a substantial probability of immediate breakdown at cycle 2. Furthermore, standard models are trained under symmetric loss functions (such as Mean Squared Error), which penalize early and late errors equally. In physical operations, this symmetry is catastrophic: predicting that an engine will survive longer than it actually does ($d > 0$) risks uncontained in-service failure, whereas predicting failure prematurely ($d < 0$) merely incurs early maintenance cost [3], [4].

To provide principled uncertainty bounds, prognostic models must emit calibrated prediction intervals $[L(x), U(x)]$ targeting nominal coverage $1 - \alpha$. While heuristic probabilistic methods (e.g., Monte Carlo dropout and Gaussian processes) rely on restrictive parametric assumptions, Conformal Prediction [5] provides a distribution-free uncertainty framework. Hierarchical conformal theory provides marginal predictive coverage under its stated sampling and exchangeability assumptions. IntelliTwin uses an engine-weighted construction motivated

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by that theory. Applicability of those assumptions to ordered, lifetime-stratified degradation trajectories is not established merely by weighting engines equally; reliability is therefore evaluated empirically under the frozen protocol.

Despite this theoretical appeal, applying conformal prediction to industrial asset prognostics reveals critical research challenges:

1) Autocorrelated Degradation Trajectories: Telemetry cycles within an engine violate observation-level exchangeability. Standard pooled conformal prediction over-weights long-lived engines, introducing survivor bias. Hierarchical, engine-weighted quantile computation [6] balances weight across engine clusters, but finite-sample reliability under trajectory autocorrelation requires empirical validation.
2) Subgroup Reliability Breakdown: Global conformal calibrators target marginal coverage averaged across all operating cycles, but frequently exhibit acute localized undercoverage during high-stress flight regimes or during accelerated terminal wear stages [7], [8].
3) The Downstream Decision Gap: Downstream sequential maintenance policies trigger replacement actions when the lower prediction bound drops below an operational threshold ($L(t) \leq 1.0$). Conformal calibration methods that treat intervals symmetrically or arbitrarily expand interval width can induce extreme operational conservatism, scrapping components prematurely under the guise of high coverage.

To bridge this gap, this paper investigates decision-aware, condition-adaptive conformal calibration for turbofan RUL prognostics. We present a research prototype framework integrating causal temporal windowing, frozen point-predictor benchmarks, exact rational hierarchical quantiles, and asymmetric nonconformity score inversion, evaluated within a discrete-event sequential maintenance simulator.

II. Related Work

A. Remaining Useful Life Point Prognostics

Prognostic health management of aero-engines was popularized by the NASA C-MAPSS dataset release [1], which formalized run-to-failure simulation across varied operational flight conditions and fault modes. Standard point predictors model RUL using deep neural networks or gradient-boosted decision trees. While the C-MAPSS asymmetric scoring metric heavily penalizes overestimation ($d > 0$), base models are routinely optimized under symmetric loss objectives. Furthermore, point predictions provide no confidence intervals, depriving operators of actionable risk metrics.

B. Conformal Uncertainty Quantification in PHM

Conformal prediction [5] has emerged as an authoritative distribution-free uncertainty framework. Conformalized Quantile Regression (CQR) was adapted to aero-engines by Diao et al. [4] using LSTM networks on C-MAPSS FD001 and FD002, and by Robinson [9] across all four C-MAPSS subsets. However, standard split conformal methods assume observation exchangeability. Because sensor telemetry is autocorrelated within an engine, pooling raw cycles violates exchangeability and over-represents long-lived engines. Lee, Barber, and Willett [6] proved that for clustered hierarchical data, finite empirical mass must be bounded by $\frac{m}{m+1}$, with the remaining $\frac{1}{m+1}$ probability mass placed at $+\infty$, using engine-weighted quantiles to guarantee finite-sample coverage for unseen clusters.

C. Conditional Reliability Diagnostics and Maintenance Economics

Recent studies emphasize that marginal coverage obscures severe conditional vulnerabilities. Yang et al. [7] audited empirical calibration across operating-regime shifts in rotating bearings, revealing acute localized undercoverage. BenAbdennour [8] demonstrated grouped validation and interval-safety calibration across degradation domains, showing non-linear expansion of uncertainty during terminal wear stages.

Concurrently, uncertainty-aware maintenance scheduling has been investigated via simulation and reinforcement learning [3], [10], [11]. Cortes-Gomez et al. [12] introduced Utility-Directed Conformal Prediction (UDCP), demonstrating that tailoring nonconformity scores to downstream decision utility substantially reduces operational loss. IntelliTwin synthesizes these domains, evaluating whether asymmetric decision-aware score inversion can improve sequential maintenance utility while preserving subgroup reliability and coverage safety.

III. Problem Formulation and Hypotheses

A. Trajectory Formulation and Dual-Horizon Targets

Let $i \in \{1, \ldots, N\}$ denote an engine unit, and let $t \in \{1, \ldots, T_i\}$ denote operational flight cycles, where T_i is the cycle of total failure. At each cycle t, an engine produces raw telemetry $x_{it} \in \mathbb{R}^{17}$ comprising 3 operating settings and 14 qualified degradation sensors.

Causal sliding windows of length $W = 30$ cycles with stride 1 and strictly no padding are extracted:

$$X_{it} = [x_{i,t-29},\ x_{i,t-28},\ \ldots,\ x_{it}] \in \mathbb{R}^{30\times 17} \tag{1}$$

Qualified inference endpoints are $V_i = \{30, 31, \ldots, T_i\}$, yielding $n_i = T_i - 29$ valid predictions.

Under ADR-012, two distinct target horizons are formalized:

$$y_{\text{cap}}(t) = \min\left(125,\ T_i - t\right) \quad \text{(Capped Training Target)} \tag{2}$$

$$y(t) = T_i - t \quad \text{(Uncapped Evaluation \& Calibration Target)} \tag{3}$$

Base models are optimized on y_{cap} to prevent overfitting on early-life sensor plateaus. Calibration nonconformity scores and downstream maintenance simulations evaluate true uncapped longevity y.

B. Conformal Prediction Intervals

Given a base predictor point estimate $p_{it} = \max(0.0,\ \hat{y}(X_{it}))$, a conformal calibrator constructs a prediction interval:

$$C(X_{it}) = [L_{it},\ U_{it}] \tag{4}$$

such that empirical marginal coverage satisfies $\mathbb{P}(y \in C(X)) \geq 1 - \alpha$. In this study, nominal coverage is set to 90% ($\alpha = 0.10$), with secondary sensitivities at 80% and 95%.

C. The Pre-Registered Hypothesis Framework (ADR-018)

The study evaluates four pre-registered hypotheses governed by the following decision logic:

- RQ1 (Maintenance Utility):

  $H_{0,1} : \Delta_{\text{cost}} \leq 0.20$ SCU or $E_{\text{coverage}} \geq 0.05$ or $\Delta_{\text{fail}} > 0.0$

  $H_{1,1} : \Delta_{\text{cost}} > 0.20$ SCU and $E_{\text{coverage}} < 0.05$ and $\Delta_{\text{fail}} \leq 0.0$

  evaluated against global_hcp and regime_hcp under primary_reference on FD002.
- RQ2 (Subgroup Reliability):

  $H_{0,2} : \Delta_{\text{error}} \leq 0.01$ or $D_{\text{width}} \geq 1.0$ or $D_{\text{score}} \geq 1.0$

  $H_{1,2} : \Delta_{\text{error}} > 0.01$ and $D_{\text{width}} < 1.0$ and $D_{\text{score}} < 1.0$

  evaluated across operating regimes (C_1–C_6) and lifecycle stages (Early/Mid/Late).
- RQ3 (Decision-Value Attribution):

  $$H_{1,3} : \Delta_{\text{cost}} > 0.20\,\text{SCU over all four prespecified matched controls} \tag{5}$$
- RQ4 (Generalization & Non-Harm):

  $$H_{1,4} : \Delta^{\text{agg}}_{\text{cost}} > 0.20\,\text{SCU},\ \Delta^{\text{agg}}_{\text{error}} > 0.01,\ \text{and no cell harm} \tag{6}$$

  evaluated across the balanced 6-cell matrix (FD001, FD002, FD004 × GBR, LSTM).

IV. Proposed Methodology

A. Data Pipeline, Preprocessing, and Leakage Controls

The empirical pipeline processes C-MAPSS subsets FD001, FD002, and FD004 (totaling 609 engines). All data partitioning is executed strictly at the independent engine cluster level: approximately 50% training, 20% calibration, 10% validation, and 20% held-out test engines (manifests preserved in data/splits/).

Preprocessing uses training-only z-score normalization. Means and population standard deviations are estimated with ddof=0. An exactly zero standard deviation is replaced by an effective scale of 1.0, while the mean is still subtracted:

$$x_{\text{normalized}} = \frac{x - \mu_{\text{train}}}{s_{\text{train}}} \tag{7}$$

where $s_{\text{train}} = \sigma_{\text{train}}$ when $\sigma_{\text{train}} > 0$, and $s_{\text{train}} = 1.0$ otherwise.

Dataset-specific processing is structured as follows:

- FD001: Single nominal operating condition; global training-fitted z-score normalization across all qualified features.
- FD002 and FD004: Operational flight settings are standardized using training data; six operating regimes (C_1–C_6) are learned from standardized training settings using K-Means clustering ($k = 6$, random_state = 2026, n_init = 10). Canonical regime labels are assigned based on lexicographic centroid ordering. Sensor normalization parameters are fitted separately within each training regime.

Preprocessing executes strictly prior to window construction. The pipeline enforces a frozen 17-feature selection supported by the recorded qualification procedure:

setting1, setting2, setting3, s2, s3, s4, s7, s8, s9, s11, s12, s13, s14, s15, s17, s20, s21

Seven sensors ($s_1, s_5, s_6, s_{10}, s_{16}, s_{18}, s_{19}$) are excluded under this recorded qualification procedure; we do not claim they are mathematically constant or universally uninformative across all datasets and regimes.

Causal temporal windowing extracts chronological sequences of length $W = 30$ cycles with stride 1, strictly no padding, and no crossing of engine boundaries. In training splits, target RUL is capped at $\min(\text{rul_true}, 125)$ cycles as a modeling convention. Point predictions are evaluated as $p = \max(0, \text{raw_prediction})$ without upper clipping at 125.

B. Frozen Base Predictor Matrix

Two base predictor families receive identical 30-cycle causal information horizons:

- Histogram Gradient Boosting Regressor (GBR): sklearn.ensemble.HistGradientBoostingRegressor operating on C-order flattened 30×17 windows (510 features). Winning configurations selected on validation data across seeds 2026–2027: gbr_lr010_leaf15_l2_0 for FD001, and gbr_lr010_leaf31_l2_0 for FD002 and FD004.
- Unidirectional LSTM: PyTorch neural network comprising one unidirectional recurrent layer (input_size=17, hidden_size=64, num_layers=1, batch_first=True, bidirectional=False, internal LSTM dropout=0.0), extracting the final sequence output, followed by a separate nn.Dropout(p=0.20) and nn.Linear(64, 1). Winning configuration: lstm_h64_lr0010 (learning rate $\eta = 0.0010$), using early stopping on validation uncapped-RUL RMSE.

Each selected configuration is trained across five final research seeds (2026, 2027, 2028, 2029, 2030), yielding 30 frozen checkpoints cataloged in experiments/base_models/model_registry.json. We do not

characterize the five GBR seeds as independent experiments, nor assert universal deterministic behavior beyond the recorded runs.

C. Hierarchical Engine-Weighted Conformal Calibration

Standard pooled conformal prediction introduces survivor bias in industrial run-to-failure telemetry because long-lived engines contribute more sliding windows than short-lived units. IntelliTwin implements hierarchical engine-weighted quantiles motivated by Lee, Barber, and Willett [6].

For FINAL calibration with m contributing engines, where engine i contains n_i valid endpoints, the empirical distribution function is:

$$F_{\text{final}}(q) = \frac{1}{m+1} \sum_{i=1}^{m} \frac{1}{n_i} \sum_{t=1}^{n_i} \mathbb{I}(\text{score}_{it} \leq q) \tag{8}$$

The mass of each finite endpoint is $\frac{1}{(m+1)n_i}$. The total finite mass is $\frac{m}{m+1}$, with the remaining probability mass $\frac{1}{m+1}$ placed at $+\infty$. The conformal threshold q^* for nominal coverage $1-\alpha$ is the smallest observed score whose CDF reaches $1-\alpha$. If no finite score reaches it, $q^* = +\infty$. (In contrast, normalized development empirical CDFs use $1/m$ weighting when selecting configurations via leave-one-engine-out cross-validation).

D. Formulations of Evaluated Conformal Methods

Let $p \geq 0$ denote non-negative predicted RUL in cycles, $y = y \geq 0$ denote uncapped ground-truth RUL, and $g \in \{C_1, \ldots, C_6\}$ denote the observed operational regime.

1) Global Hierarchical Conformal Prediction (global_hcp): Uses symmetric absolute residuals $\text{score}(p, y) = |y - p|$, fitting a single global threshold q_{global}, yielding $L = \max(0, p - q_{\text{global}})$ and $U = p + q_{\text{global}}$.

2) Condition-Aware Hierarchical Conformal Prediction (regime_hcp): Uses symmetric absolute residuals, fitting separate regime-specific thresholds q_g. Regime support requires $m_g \geq \max(10, \lceil 1/\alpha \rceil - 1)$ distinct contributing engines (10 engines for 80% and 90% coverage; 19 engines for 95% coverage). If m_g is insufficient, the method falls back deterministically to q_{global} fitted for the same method and coverage level, recording fallback.

3) Proposed Decision-Aware Asymmetric Conformal Prediction (decision_aware_hcp): Modulates interval width and asymmetry via scaling functions:

$$\sigma(p) = \left(1 + \frac{p}{125}\right)^{\gamma}, \quad z(p) = \exp\left(-\frac{p}{20}\right) \tag{9}$$

$$a(p) = \sigma(p)\exp(\beta z(p)), \quad b(p) = \sigma(p)\exp(-\beta z(p)) \tag{10}$$

where $\gamma \in \{0, 1, 2\}$ and $\beta \in \{-\ln 2, 0, +\ln 2\}$. Nonconformity scores and inverted intervals are:

$$\text{score}(p, y) = \max\left(\frac{p-y}{a(p)},\ \frac{y-p}{b(p)},\ 0\right) \tag{11}$$

$$L = \max(0,\ p - q_g \cdot a(p)), \quad U = p + q_g \cdot b(p) \tag{12}$$

E. Mathematical Mechanics of Negative Beta ($\beta < 0$)

At fixed p, q, and γ, negative β decreases $a(p)$ relative to $\beta = 0$ and increases $b(p)$. The lower bound L is nondecreasing and the upper bound U is nondecreasing. We do not claim $a < 1$ or $b > 1$ for every p, as the $\sigma(p)$ factor also influences their magnitude. Furthermore, this mathematical fact does not establish improved maintenance utility or earlier intervention; nonconformity thresholds q^* are fitted separately and utility must be evaluated empirically in sequential simulation.

Candidate gamma2_betam1 ($\gamma = 2, \beta = -\ln 2$) was selected as the optimal feasible configuration across all six mandatory cells and frozen under ADR-021.

V. Experimental Setup

A. Benchmark Datasets and Engine-Level Partitions

We evaluate IntelliTwin on three benchmark subdatasets from the NASA C-MAPSS turbofan engine degradation simulation suite [1]:

1) FD001: Single sea-level operating condition, single failure mode (HPC degradation), comprising 100 run-to-failure engine trajectories.
2) FD002: Six diverse operating flight conditions, single failure mode (HPC degradation), comprising 260 engine trajectories.
3) FD004: Six diverse operating flight conditions, two concurrent failure modes (HPC and fan degradation), comprising 249 engine trajectories.

To prevent temporal and engine-level data leakage, partitioning is strictly executed at the engine level using a fixed master seed (2026) prior to any data preprocessing (50% training, 20% calibration, 10% validation, 20% held-out test):

- FD001 (100 engines): 50 training, 20 calibration, 10 validation, and 20 held-out test engines.
- FD002 (260 engines): 130 training, 52 calibration, 26 validation, and 52 held-out test engines, stratified across engine lifetime quartiles (Q_1–Q_4).
- FD004 (249 engines): 124 training, 50 calibration, 25 validation, and 50 held-out test engines.

Across all three datasets, 304 engines are allocated to training, 122 to calibration, 61 to validation, and 122 to final held-out testing (total 609 engines).

B. Feature Space and Causal Windowing

The raw engine streams comprise 3 operational settings and 21 sensor measurements per cycle. Preprocessing applies training-only z-score normalization ($x_{\text{normalized}} = (x - \mu_{\text{train}})/s_{\text{train}}$, with ddof = 0 and $s = 1.0$ if $\sigma = 0$). A locked 17-feature selection is enforced: 3 operational settings and 14 qualified degradation sensors. Seven sensors are excluded under the recorded qualification procedure; we do not claim they are universally uninformative across all domains.

Temporal trajectories are windowed with an exact causal horizon of $w = 30$ cycles, unit stride ($s = 1$), and strict avoidance of padding or crossing engine boundaries. Training targets use the modeling convention $\min(\text{rul_true}, 125)$ cycles, while calibration, validation, and test splits evaluate true uncapped RUL.

C. Base Degradation Predictors

Two point-prediction architectures operate on identical 30-cycle causal horizons:

1) Gradient Boosted Regressor (GBR): sklearn.ensemble.HistGradientBoostingRegressor operating on C-order flattened $30 \times 17 = 510$-dimensional vectors. Selected configs: gbr_lr010_leaf15_l2_0 (FD001) and gbr_lr010_leaf31_l2_0 (FD002, FD004).
2) Long Short-Term Memory (LSTM): Single-layer unidirectional PyTorch LSTM ($H = 64$, recurrent dropout 0.0, followed by nn.Dropout(p=0.20) and nn.Linear(64, 1), learning rate $\eta = 0.0010$, lstm_h64_lr0010) with early stopping on validation uncapped RMSE.

Each configuration is trained across five random seeds $(2026, 2027, 2028, 2029, 2030)$, producing 30 frozen base model checkpoints in the registry.

D. Calibration Variants and Pre-Registered Controls

On the reserved calibration engine splits, we evaluate three conformal calibration variants: global_hcp, regime_hcp, and decision_aware_hcp.

To rigorously evaluate RQ3, four named controls across three distinct baseline families are pre-registered:

1) Matched-Width Interval (matched_width_interval): Sources from global_hcp or regime_hcp, scaling unclipped half-width by $\lambda \in [0.01, 4.0]$ (plus $\{5, 6, 8, 12, 16\}$): $L_\lambda = \max(0, p - \lambda q_{\text{source}}), U_\lambda = p + \lambda q_{\text{source}}$. Selected on FD002 validation data to match proposed validation MPIW within ≤ 1.0 cycle: GBR selects global_hcp with $\lambda = 0.92$; LSTM selects regime_hcp with $\lambda = 0.90$.
2) Widened Interval (widened_interval): Same scaling formula with $\lambda \geq 1.00$, minimizing validation cost: GBR selects regime_hcp with $\lambda = 1.00$; LSTM selects global_hcp with $\lambda = 1.00$.
3) Point Safety Margin (point_safety_margin): Maintains iff $p \leq 1 + m$ for $m \in \{0, \ldots, 250\}$, minimizing validation cost: GBR selects $m = 5$; LSTM selects $m = 4$. As a point policy, width and conformal coverage are not applicable (NOT_APPLICABLE).
4) Earlier Intervention (earlier_intervention): Maintains iff $L_{\text{source}} \leq h$ for $h \in \{1, \ldots, 251\}$, minimizing validation cost: GBR selects regime_hcp with $h = 1$; LSTM selects global_hcp with $h = 1$.

E. Maintenance Simulation and Cost Scenarios

Sequential policy execution is evaluated using the discrete-event simulator. The primary economic scenario (primary_reference) specifies:

$$C_{\text{PM}} = 1.0\,\text{SCU}, \quad C_{\text{FAIL}} = 5.0\,\text{SCU}, \quad C_{\text{WASTE}} = 0.20\,\text{SCU/cycle} \tag{13}$$

Sensitivity scenarios evaluate failure_sensitive ($C_{\text{FAIL}} = 10.0$ SCU) and waste_sensitive ($C_{\text{WASTE}} = 0.40$ SCU/cycle).

The primary policy triggers MAINTAIN_NOW at cycle $t \in [30, T-1]$ if and only if $L(t) \leq 1.0$ cycle. Preventive maintenance occurs immediately upon decision and terminates the episode (zero intervention lead time). If maintenance occurs at $t < T$, cost is $C_{\text{PM}} + C_{\text{WASTE}} \times (T - t)$; if no maintenance occurs before failure ($t = T$), cost is C_{FAIL} with wasted life $W = 0$. Offline interval evaluation includes all qualified endpoints 30 through T. The equality values (20, 45, and 10 cycles) compare preventive-plus-waste cost with failure cost, and do not represent policy decision thresholds.

F. Statistical Inference and Decision Protocol

Hypothesis testing uses paired engine-level cluster bootstrap inference:

- Resampling Engine: $B = 10{,}000$ paired engine-cluster bootstrap resamples with replacement, stratified by frozen engine lifetime quartiles (Q_1–Q_4), using PCG64 PRNG with seed sequence SeedSequence([2026, 6, dataset_number]).
- Weighting: Engines receive equal weighting; seeds are averaged within engines as descriptive training variability, not independent sampling units.
- Missing-Group Bound Propagation: If an eligible group (≥ 10 engines) is absent in a bootstrap resample, its absolute coverage error is bounded by $[0.0, 0.90]$ at 90% nominal coverage and propagated conservatively.
- Confirmatory Decision Intervals: 98.75% percentile intervals (quantiles 0.00625 and 0.99375), providing Bonferroni family-wise error rate control ($\alpha = 0.05/4 = 0.0125$) across RQ1–RQ4.
- Intersection-Union Test (IUT): Benefit conditions pass if lower CI $>$ margin; ceiling/noninferiority conditions pass if upper CI $<$ ceiling. Equality with a boundary or degenerate inference is inconclusive. Failure safeguard uses observed fleet failure rates ($\Delta_{\text{fail}} \leq 0.0$). An RQ is Supported only if all required components pass, Unsupported if any fails, and otherwise Inconclusive.

VI. Results and Confirmatory Evaluation

Status Notice: The empirical evaluations reported in this section represent the pre-registered reporting templates for the confirmatory evaluation. In accordance with the project's evidence

protocol, all held-out test split evaluations remain blinded during active software development and verification. Empirical entries are designated PENDING awaiting completion of experimental gates UPD-02, UPD-03, and UPD-04. No synthetic or unverified empirical numbers are reported.

A. Base Degradation Prediction Performance

Point predictive performance is evaluated across the six experimental cells on the held-out test engine splits. Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) are computed with equal engine weighting across all test engines. Table I outlines the reporting structure.

TABLE I
Point Prediction Accuracy on Held-Out Test Engines (Mean ± Std across 5 Seeds)

Dataset	Architecture	RMSE (cycles)	MAE (cycles)
FD001	GBR	PENDING (UPD-02)	PENDING (UPD-02)
FD001	LSTM	PENDING (UPD-02)	PENDING (UPD-02)
FD002	GBR	PENDING (UPD-02)	PENDING (UPD-02)
FD002	LSTM	PENDING (UPD-02)	PENDING (UPD-02)
FD004	GBR	PENDING (UPD-02)	PENDING (UPD-02)
FD004	LSTM	PENDING (UPD-02)	PENDING (UPD-02)

B. Marginal and Conditional Interval Calibration

Conformal prediction intervals are evaluated at nominal confidence level $1 - \alpha = 0.90$ across all 3 datasets $\times$ 2 models $\times$ 3 registered calibration methods (18 cells). Metrics comprise engine-weighted empirical coverage (Cov), mean prediction interval width ($\bar{W}$), and Winkler interval score ($\mathrm{IS}_{0.10}$). Table II summarizes the complete registered evaluation matrix.

C. Sequential Maintenance Decision Outcomes

Downstream maintenance simulation evaluates policies across the 18 primary method cells plus 8 pre-registered control configurations on FD002 (total 26 evaluation rows) under the frozen primary_reference scenario ($C_{\mathrm{PM}} = 1.0$ SCU, $C_{\mathrm{FAIL}} = 5.0$ SCU, $C_{\mathrm{WASTE}} = 0.20$ SCU/cycle). Policy triggers replacement at cycle $t \in [30, T-1]$ iff $L(t) \leq 1.0$ cycle. Table III reports mean cost per engine ($\bar{C}$), fleet failure rate (P_{fail}), and mean wasted life ($\bar{W}$).

D. Hypothesis Decision Outcomes (RQ1–RQ4)

Confirmatory statistical inference is governed by 10,000 paired engine-cluster bootstrap resamples with Bonferroni-corrected 98.75% confidence intervals. Table IV presents the formal decision matrix under the pre-registered Intersection-Union Test logic (ADR-018).

VII. Discussion

A. Methodological Implications for Turbofan Prognostics

The core inquiry of the IntelliTwin framework is whether distribution-free uncertainty quantification provides measurable, operational value to industrial decision-makers beyond standard point prognostics or ad-hoc safety margins. Traditional prognostic paradigms focus almost exclusively on minimizing cycle-level error metrics (e.g., RMSE or asymmetric scoring functions). However, in mission-critical applications such as aviation gas turbine maintenance, cycle-level accuracy does not translate directly into operational cost efficiency. A maintenance policy driven by an uncalibrated point predictor is structurally vulnerable to overestimation errors, where an engine is allowed to operate beyond its physical failure boundary, incurring catastrophic unplanned replacement penalties.

By establishing finite-sample marginal coverage guarantees via split-conformal inference, IntelliTwin provides decision-makers with a mathematically principled lower bound $L_t(\alpha)$. Nevertheless, conformal guarantees hold marginally across the calibration exchangeability distribution. In sequential degradation processes, temporal autocorrelation and operating regime transitions can cause local conditional coverage to deviate significantly from the nominal $1 - \alpha$ rate.

B. Conservatism, Asymmetric Penalties, and Negative β

A critical methodological focus of this work is the formalization of asymmetric, decision-aware conformal calibration. In maintenance scheduling, the economic structure reflects synthetic preferences under the primary reference scenario ($C_{\mathrm{PM}} = 1.0$ SCU, $C_{\mathrm{FAIL}} = 5.0$ SCU, $C_{\mathrm{WASTE}} = 0.20$ SCU/cycle).

The scaling functions are defined by:

$$\sigma(p) = \left(1 + \frac{p}{125}\right)^{\gamma}, \quad z(p) = \exp\left(-\frac{p}{20}\right) \tag{14}$$

$$a(p) = \sigma(p)\exp(\beta z(p)), \quad b(p) = \sigma(p)\exp(-\beta z(p)) \tag{15}$$

At fixed p, q, and γ, negative β decreases $a(p)$ relative to $\beta = 0$ and increases $b(p)$. The lower bound $L = \max(0, p - q_g a(p))$ is nondecreasing and the upper bound $U = p + q_g b(p)$ is nondecreasing. We do not claim $a < 1$ or $b > 1$ for every p, as the $\sigma(p)$ factor also influences their magnitude. Furthermore, this mathematical fact does not establish improved maintenance utility or earlier intervention; nonconformity thresholds q^* are fitted separately and utility must be evaluated empirically in sequential simulation.

C. Regime Stratification and Subgroup Reliability

In complex operational environments such as C-MAPSS FD002 and FD004, turbofans traverse six distinct operating flight regimes characterized by substantial variations in altitude, Mach number, and throttle resolver angle. Global conformal calibration aggregates residuals across

TABLE II
Primary Prediction Interval Calibration Performance at Nominal 90% Confidence Level on Held-Out Test Splits ($3 \times 2 \times 3 = 18$ cells)

Dataset	Base Model	Calibration Method	Empirical Cov. (%)	Mean Width (cycles)	Winkler Score
FD001	GBR	Global HCP (global_hcp)	PENDING	PENDING	PENDING
FD001	GBR	Regime HCP (regime_hcp)*	PENDING	PENDING	PENDING
FD001	GBR	Decision-Aware HCP (decision_aware_hcp)	PENDING	PENDING	PENDING
FD001	LSTM	Global HCP (global_hcp)	PENDING	PENDING	PENDING
FD001	LSTM	Regime HCP (regime_hcp)*	PENDING	PENDING	PENDING
FD001	LSTM	Decision-Aware HCP (decision_aware_hcp)	PENDING	PENDING	PENDING
FD002	GBR	Global HCP (global_hcp)	PENDING	PENDING	PENDING
FD002	GBR	Regime HCP (regime_hcp)	PENDING	PENDING	PENDING
FD002	GBR	Decision-Aware HCP (decision_aware_hcp)	PENDING	PENDING	PENDING
FD002	LSTM	Global HCP (global_hcp)	PENDING	PENDING	PENDING
FD002	LSTM	Regime HCP (regime_hcp)	PENDING	PENDING	PENDING
FD002	LSTM	Decision-Aware HCP (decision_aware_hcp)	PENDING	PENDING	PENDING
FD004	GBR	Global HCP (global_hcp)	PENDING	PENDING	PENDING
FD004	GBR	Regime HCP (regime_hcp)	PENDING	PENDING	PENDING
FD004	GBR	Decision-Aware HCP (decision_aware_hcp)	PENDING	PENDING	PENDING
FD004	LSTM	Global HCP (global_hcp)	PENDING	PENDING	PENDING
FD004	LSTM	Regime HCP (regime_hcp)	PENDING	PENDING	PENDING
FD004	LSTM	Decision-Aware HCP (decision_aware_hcp)	PENDING	PENDING	PENDING

*On FD001 (single operating condition), regime_hcp operates with a single regime partition identical to global calibration.

TABLE III
Sequential Maintenance Policy Outcomes on Held-Out Test Engines (primary_reference: $C_{\text{PM}} = 1.0, C_{\text{FAIL}} = 5.0, C_{\text{WASTE}} = 0.20$ SCU; 26 rows)

Dataset	Model	Policy / Calibration Strategy	Cost / Engine (SCU)	Failures (%)	Wasted Life (cycles)
Panel A: Primary Registered Calibration Methods (18 Cells)					
FD001	GBR	Global HCP Lower Bound (global_hcp)	PENDING	PENDING	PENDING
FD001	GBR	Regime HCP Lower Bound (regime_hcp)	PENDING	PENDING	PENDING
FD001	GBR	Decision-Aware Lower Bound (decision_aware_hcp)	PENDING	PENDING	PENDING
FD001	LSTM	Global HCP Lower Bound (global_hcp)	PENDING	PENDING	PENDING
FD001	LSTM	Regime HCP Lower Bound (regime_hcp)	PENDING	PENDING	PENDING
FD001	LSTM	Decision-Aware Lower Bound (decision_aware_hcp)	PENDING	PENDING	PENDING
FD002	GBR	Global HCP Lower Bound (global_hcp)	PENDING	PENDING	PENDING
FD002	GBR	Regime HCP Lower Bound (regime_hcp)	PENDING	PENDING	PENDING
FD002	GBR	Decision-Aware Lower Bound (decision_aware_hcp)	PENDING	PENDING	PENDING
FD002	LSTM	Global HCP Lower Bound (global_hcp)	PENDING	PENDING	PENDING
FD002	LSTM	Regime HCP Lower Bound (regime_hcp)	PENDING	PENDING	PENDING
FD002	LSTM	Decision-Aware Lower Bound (decision_aware_hcp)	PENDING	PENDING	PENDING
FD004	GBR	Global HCP Lower Bound (global_hcp)	PENDING	PENDING	PENDING
FD004	GBR	Regime HCP Lower Bound (regime_hcp)	PENDING	PENDING	PENDING
FD004	GBR	Decision-Aware Lower Bound (decision_aware_hcp)	PENDING	PENDING	PENDING
FD004	LSTM	Global HCP Lower Bound (global_hcp)	PENDING	PENDING	PENDING
FD004	LSTM	Regime HCP Lower Bound (regime_hcp)	PENDING	PENDING	PENDING
FD004	LSTM	Decision-Aware Lower Bound (decision_aware_hcp)	PENDING	PENDING	PENDING
Panel B: Pre-Registered Matched Controls on FD002 (8 Cells)					
FD002	GBR	Matched-Width Interval (global_hcp, $\lambda = 0.92$)	PENDING	PENDING	PENDING
FD002	GBR	Widened Interval (regime_hcp, $\lambda = 1.00$)	PENDING	PENDING	PENDING
FD002	GBR	Point Safety Margin ($p \leq 1 + 5$)	PENDING	PENDING	PENDING
FD002	GBR	Earlier Intervention (regime_hcp, $h = 1$)	PENDING	PENDING	PENDING
FD002	LSTM	Matched-Width Interval (regime_hcp, $\lambda = 0.90$)	PENDING	PENDING	PENDING
FD002	LSTM	Widened Interval (global_hcp, $\lambda = 1.00$)	PENDING	PENDING	PENDING
FD002	LSTM	Point Safety Margin ($p \leq 1 + 4$)	PENDING	PENDING	PENDING
FD002	LSTM	Earlier Intervention (global_hcp, $h = 1$)	PENDING	PENDING	PENDING

For point_safety_margin, prediction interval width and conformal coverage metrics are designated NOT_APPLICABLE.

TABLE IV
Pre-Registered Confirmatory Hypothesis Decision Rules and Outcome Contrasts (ADR-018)

Research Question	Contrast Metric & Scope	Pre-Registered Thresholds (98.75% CI)	Point Est.	98.75% CI	Status
RQ1 (Utility & Coverage)	Cost improvement vs. global_hcp and regime_hcp on FD002 (GBR and LSTM)	$\Delta_{\text{cost}} > 0.20$ SCU; coverage errors < 0.05; $\Delta_{\text{fail}} \leq 0.0$	PENDING	PENDING	PENDING
RQ2 (Subgroup Reliability)	Worst-group coverage error contrast for regimes (C_1–C_6) and stages on FD002	$\Delta_{\text{error}} > 0.01$; worst error < 0.05; $D_{\text{width}} < 1.0$; $D_{\text{score}} < 1.0$ cycle	PENDING	PENDING	PENDING
RQ3 (Control Attribution)	Cost improvement vs. each of the 4 matched controls on FD002 (GBR and LSTM)	$\Delta_{\text{cost}} > 0.20$ SCU over every control; coverage < 0.05; $\Delta_{\text{fail}} \leq 0.0$	PENDING	PENDING	PENDING
RQ4 (Generalization)	Balanced 6-cell evaluation matrix ($w_k = 1/6$ each) across FD001, FD002, FD004	Aggregate: $\Delta_{\text{cost}} > 0.20, \Delta_{\text{error}} > 0.01$; Cell non-harm: $\Delta_{\text{cost}} > -0.20, \Delta_{\text{error}} > -0.01$	PENDING	PENDING	PENDING

Decision rules: An RQ is decided as Supported iff all required component conditions pass at the 98.75% level, Unsupported if any condition fails, and otherwise Inconclusive.

all regimes, which can mask severe local undercoverage in rare or highly stressed operating conditions.

Regime-stratified HCP explicitly conditions nonconformity scores on the active flight regime cluster $g \in \{C_1, \ldots, C_6\}$. Regime support requires at least $m_g \geq \max(10, \lceil 1/\alpha \rceil - 1)$ distinct contributing calibration engines in that regime (10 engines for 80% and 90% coverage; 19 engines for 95% coverage). When a regime contains fewer than this threshold, the framework executes a deterministic fallback to the pooled global nonconformity quantile for the same method and coverage level, recording fallback.

D. Interpreting Potential Hypothesis Outcomes

The pre-registered statistical protocol explicitly defines decision rules for three potential outcomes across all four research questions:

- Supported: All required component conditions pass at the 98.75% decision interval level, demonstrating statistically significant and practically meaningful superiority.
- Inconclusive: No condition fails, but at least one decision interval spans the practical threshold boundary, or inference is degenerate. In this case, no definitive superiority claim is made, and the result is cataloged as requiring expanded engine sample sizes.
- Unsupported: Any required component condition fails (e.g., upper CI below margin, lower CI above ceiling, or fleet failure rate increases). This outcome indicates that the pre-registered criteria were not supported under the specified experiment.

By enforcing strict pre-registered boundaries and reporting all comparisons transparently, the research framework ensures that conclusions are dictated entirely by empirical evidence. Final empirical conclusions remain pending held-out evaluation.

VIII. Conclusion and Future Work

A. Summary of Methodological Framework

In this investigation, we formulated and pre-registered the IntelliTwin prognostic and predictive maintenance framework. Designed to address the critical gap between cycle-level machine learning accuracy and downstream operational cost in industrial turbofan fleets, IntelliTwin couples gradient boosted and recurrent deep learning base predictors with hierarchical, decision-aware conformal calibration.

The framework enforces rigorous methodological safeguards to preserve empirical validity:

1) Strict Leakage Isolation: Complete engine-level disjoint partitioning (train, calibration, validation, held-out test) executed prior to feature scaling or causal windowing ($w = 30$, stride 1, no padding).
2) Equal Engine Weighting: Evaluation metrics and conformal quantiles are computed with equal weighting per engine trajectory, eliminating length bias inherent in pooled cycle-level evaluations.
3) Principled Baseline Controls: Evaluation against four named control strategies across three distinct families (matched-width heuristic intervals, widened intervals, point safety margins, and earlier fixed interventions) to isolate the true economic value of calibrated uncertainty.
4) Pre-Registered Statistical Inference: Family-wise error rate control via Bonferroni-corrected 98.75% confidence intervals generated through 10,000 paired engine-level cluster bootstrap resamples.

B. Current Status and Verification Gates

At the current stage of project execution (pinned to commit bb9e34579343881a831e3f052608ee15e183df76), the theoretical formulation, data pipelines, base model

training protocols, and statistical evaluation machinery have been fully specified, audited, and frozen. Pretest verification artifacts (step08_full_audit.json and pretest_audit.json) confirm that all 60 base-model validation and calibration partition evaluations reproduce with zero maximum difference (0.0), establishing PRETEST_READY status. The baseline inventory comprises 652 tracked, protected Git artifacts and 72 local-only historical search checkpoints.

The software is an academic research prototype evaluating an offline, trace-driven simulation replay engine; it makes no claim of production readiness, aircraft certification, physical hardware integration, or independently established end-to-end operational reproducibility. Empirical results on held-out test splits remain blinded and NOT_STARTED, with confirmatory research question outcomes PENDING. Upon completion of official held-out evaluation, verified empirical results will be incorporated into this document.

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